

Bachelor in Artificial Intelligence

An AI-Based Decision Support System for Retinal OCT Classification: Comparing Deep Learning and Foundation Models

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Clinical Background & Motivation

Optical Coherence Tomography (OCT)

- Low-coherence interferometry providing μm -scale cross-sectional resolution
- ~30 million OCT scans performed worldwide every year

AMD

Age-Related Macular
Degeneration

Subretinal fluid & neovascular membrane

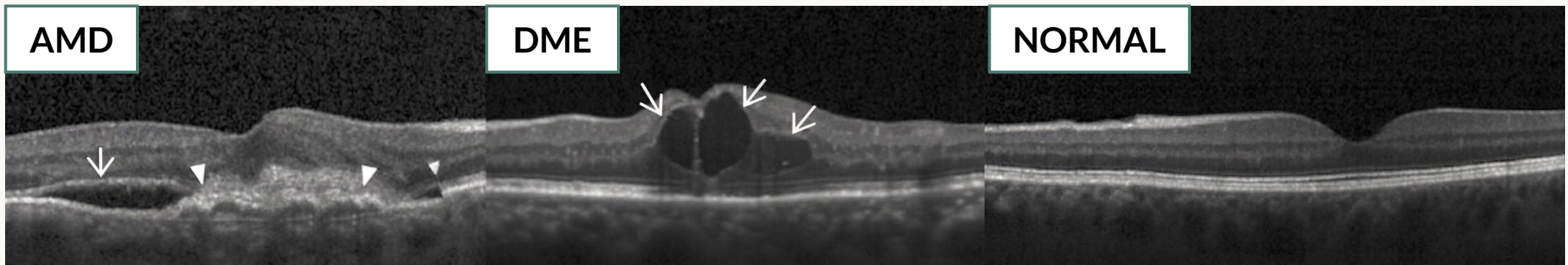
170M patients today → 288M by 2040

DME

Diabetic Macular Edema

Intraretinal cystoid spaces & retinal thickening

530M+ diabetics today → ~800M by 2045



Motivation & Objectives

Traditional Deep Learning

- Custom CNN (Lightweight Baseline)
- EfficientNet-B0 (General Transfer Learning)



Medical Foundation Models

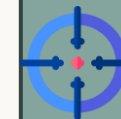
- MedGemma-4B (Multimodal Backbone)
- MIRAGE-Base (Domain-Specific ViT)

1. Accuracy

Predictive performance on diagnostic classes.

2. Calibration

Alignment of model confidence with true accuracy.



3. OOD Detection

Safely handling unseen anomalies (e.g., Glaucoma).

4. Explainability

Spatial validation via saliency and occlusion maps.

Dataset and Patient-Level Splitting

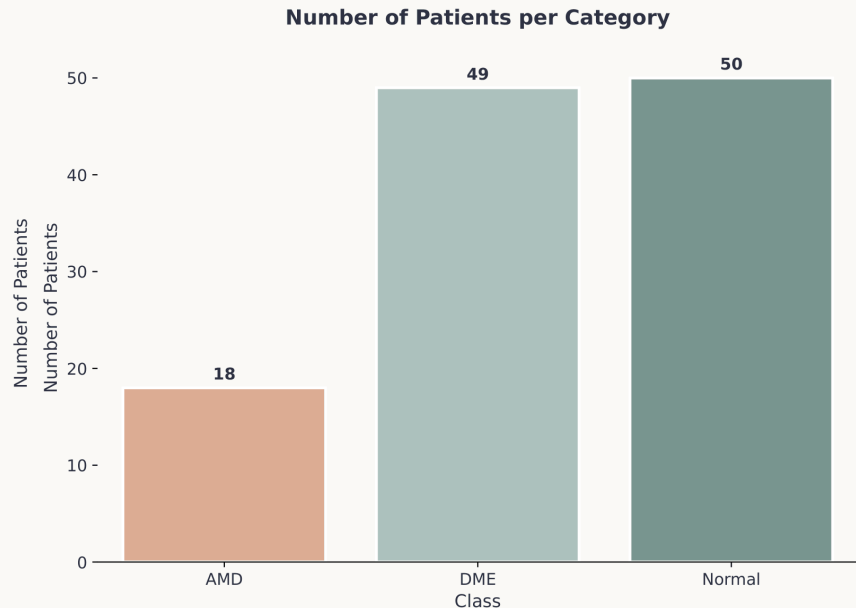
OCT5k from UCL

3,231
OCT B-scans

117
Patients

3
Classes

Initial dataset: 3,348 images - 117 fundus photos removed



Class Imbalance:

Normal cases (~50%) vs. AMD (~15%).

Addressed robustly using **class-weighted loss functions** during training.

Patient-Aware Split – StratifiedGroupKFold

Train

~66% 77 patients

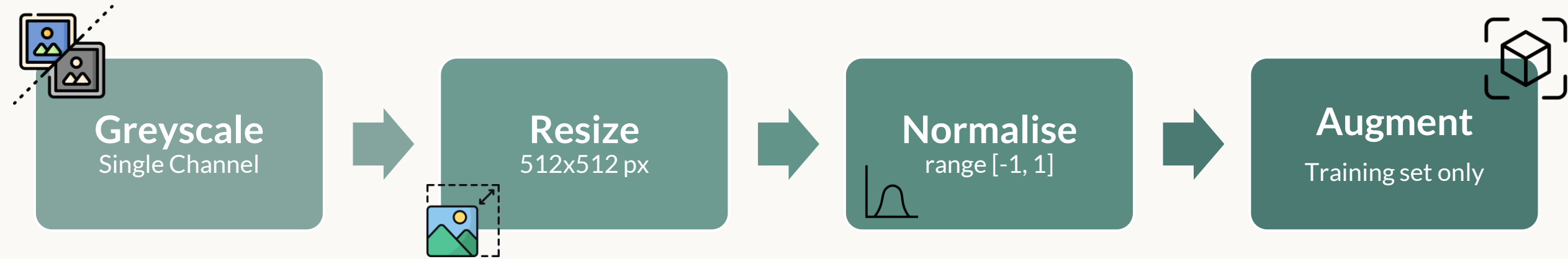
Validation

~18% 21 patients

Test

~16% 19 patients

Preprocessing



Applied identically across training, validation, and test sets (augmentation excluded for val/test).

The Four Models

Custom CNN

< 95,000 parameters; no pretraining

- 3 conv blocks: Conv + BatchNorm + ReLU + MaxPool
- Fully-connected head with dropout, trained from scratch
- Adam optimiser, lr = 1e-3, weight decay 1e-4
- Baseline for the comparison

EfficientNet-B0

≈ 4M parameters; ImageNet pretrained

- Two-stage training: linear probing → full fine-tuning
- Phase 1 (epochs 1–10): backbone frozen, head only
- Phase 2: backbone unfrozen, learning rate reduced
- Adam optimiser + ReduceLROnPlateau scheduler

MedGemma-4B

MedSigLIP vision encoder; frozen throughout

- Medical vision-language foundation model (Google)
- Encoder used purely as a frozen feature extractor
- MLP head: Linear → LayerNorm → GELU → Dropout
- Adam optimiser, lr = 3e-4, up to 80 epochs

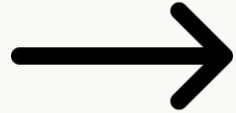
MIRAGE-Base

Pretrained on 261k OCT; SLO pairs, 42k patients

- Vision Transformer, retina-specific foundation model
- Phase 1: linear probing (backbone frozen)
- Phase 2: full fine-tuning, lr = 1e-5
- AdamW optimiser, cosine annealing scheduler

Evaluation

Accuracy



Is the prediction correct?

Calibration



Does confidence match reality?

OOD Detection



Does it know what it doesn't know?

Explainability



Can we see why it decided?

Results: Classification Performance

85.4%

CustomCNN

95.7%

EfficientNet-B0

90.0%

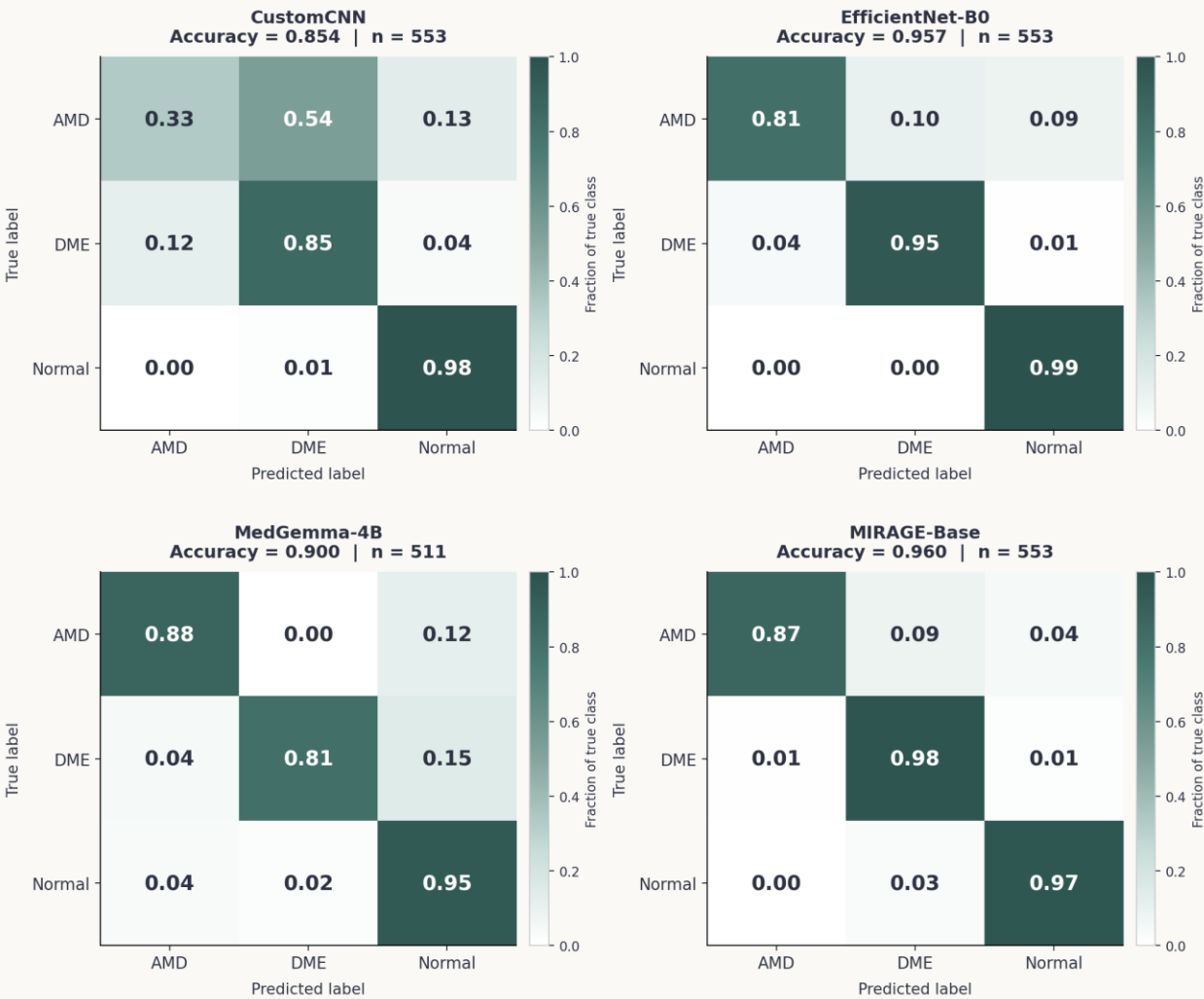
MedGemma-4B

96.0%

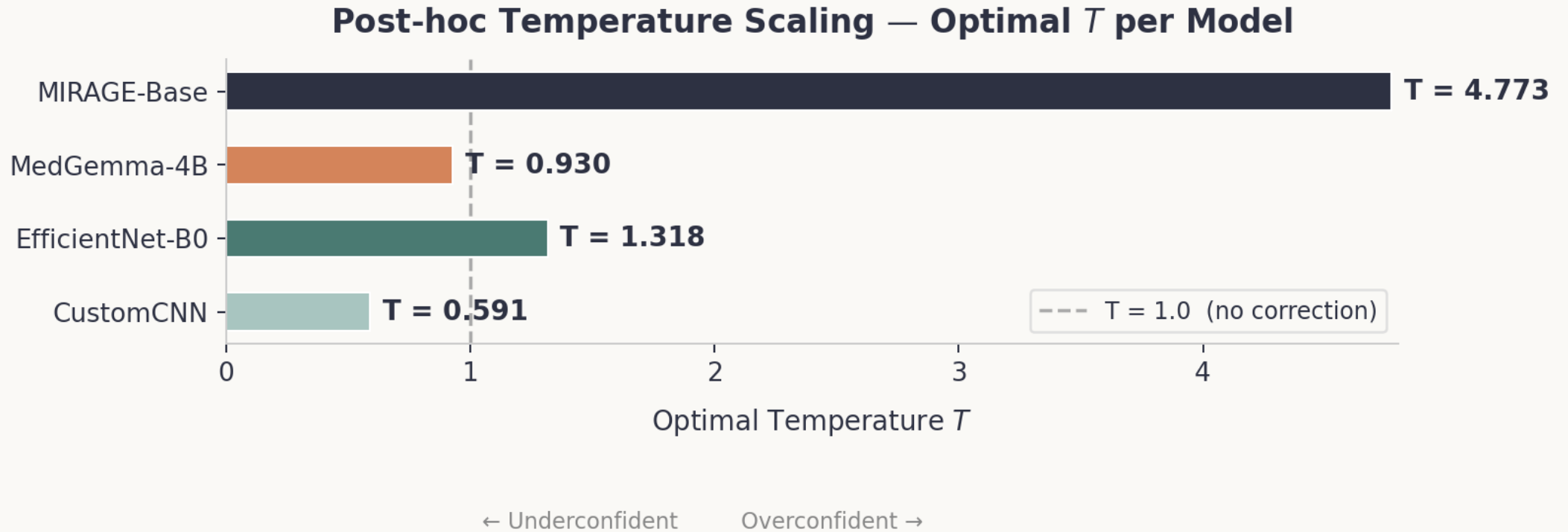
MIRAGE-Base

Model	Precision	Recall	F1-score	AUC
CustomCNN	0.747	0.722	0.729	0.917
EfficientNet-B0	0.932	0.920	0.926	0.995
MedGemma-4B	0.886	0.879	0.880	0.979
MIRAGE-Base	0.960	0.939	0.949	0.995

Normalised Confusion Matrices — Held-out Test Set



Results: Calibration

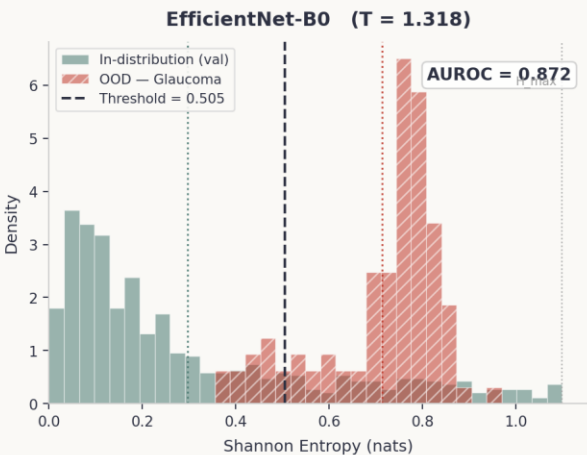
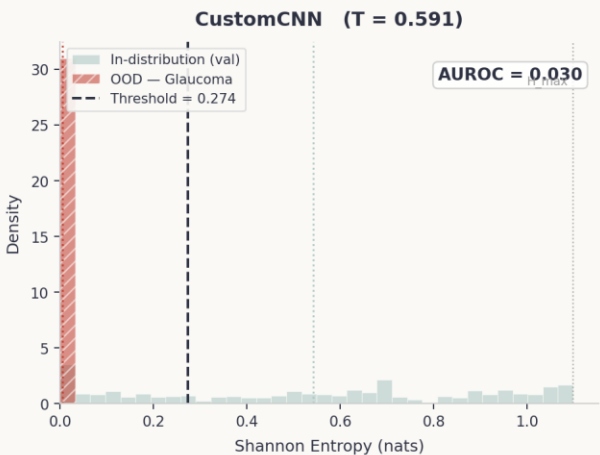


MIRAGE-Base: the most accurate model is also the most overconfident

Results: Out-of-Distribution Detection

Glaucoma: a pathology never seen in training was shown to all four models

Shannon Entropy Distributions — In-Distribution vs OOD Glaucoma
After Temperature Scaling

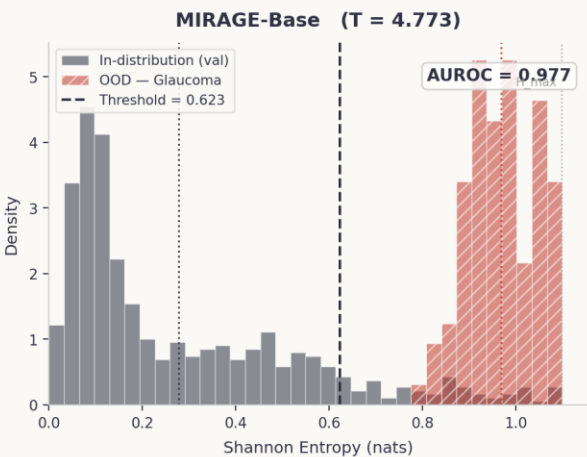
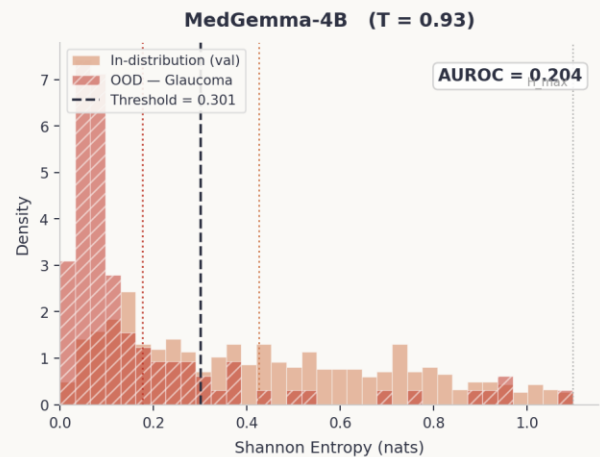


0.030
CustomCNN · AUROC

0.204
MedGemma-4B · AUROC

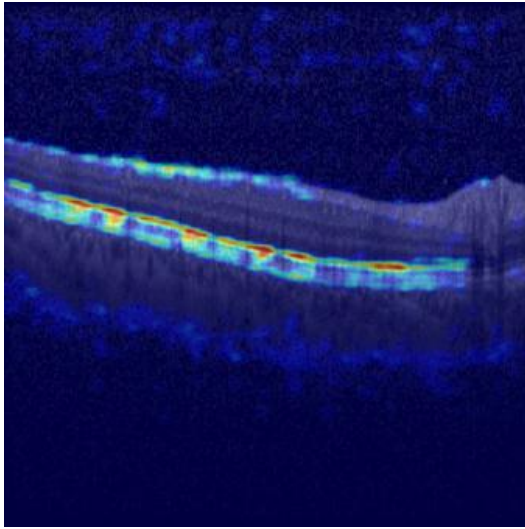
0.872
EfficientNet-B0 · AUROC

0.977
MIRAGE-Base · AUROC



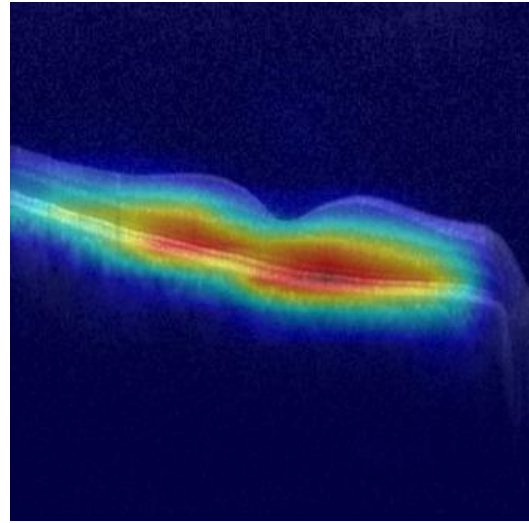
Model	Flagged (Before)	Flagged (After)
CustomCNN	0/100	0/100
EfficientNet-B0	42/100	87/100
MedGemma-4B	7/100	7/100
MIRAGE-Base	26/100	96/100

Results: Explainability



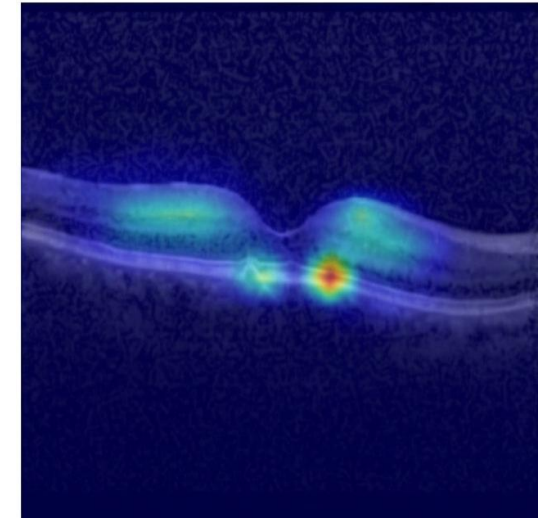
CustomCNN

Grad-CAM → diffuse attention



EfficientNet-B0

Grad-CAM → anatomically focused



MIRAGE-Base

*Occlusion sensitivity
(Vision Transformer Architecture)*

MedGemma-4B: not explainable → frozen embedding leaves no spatial signal to visualise

Comparative Discussion

	Accuracy	Calibration	OOD Safety	Explainability
CustomCNN	Weak	Mixed	Weak	Mixed
EfficientNet-B0	Strong	Mixed	Strong	Strong
MedGemma-4B	Strong	Strong	Weak	Weak
MIRAGE-Base	Strong	Weak	Strong	Strong

 Strong  Mixed  Weak

EfficientNet-B0 offers the most balanced overall profile.

The ORCA (OCT Retinal Classification Assistant) Prototype



Conclusions

01

Trustworthiness has four axes

Accuracy, calibration, OOD safety and explainability were evaluated independently.

02

No single model wins everywhere

MIRAGE-Base leads on accuracy and OOD detection (AUROC 0.977) but needs the largest calibration correction, $T = 4.773$.

03

EfficientNet-B0

The best balance of accuracy, calibration, safety and Grad-CAM explainability.

04

Known limitations

No demographic metadata to test fairness, and possible undisclosed pretraining overlap for MIRAGE.



Thank you for your attention!

Questions & Discussion